ESP32-S3 Audio Device: Development Log

August 25 – 31, 2026

1. Progress Timeline

Date	Work Completed
Aug 25	Set up PlatformIO and spent most of the day troubleshooting it. Installed the CH343 driver (wrong driver). Reviewed ESP32-S3 DevKit specs, boot control and restricted pins.
Aug 26	Installed the correct driver (FTDI VCP). Built and uploaded first test program: on-board LED gradually changing color.
Aug 29	Studied the MAX98357A pinout and the ESP32-S3 GPIO Matrix. Wired the amplifier to the board; uploaded code, but no sound.
Aug 30	Studied I2S and PCM audio. Fixed the silent amplifier (wiring issue). Speaker now produces a loud test tone.
Aug 31	Recorded and converted audio clips in Audacity. Set up SoftAP mode and a web server so clips can be triggered from a phone browser.

2. Debugging & Troubleshooting

Issue	Cause	Resolution
Board not working with PC / PlatformIO	CH343 driver was the wrong one; forcing it on crashed the computer.	Installed the FTDI VCP driver.
MAX98357A silent after wiring and upload	Poor connection: wires too thin and soldering not solid.	Switched to thicker wires and re-checked solder joints.
Audio clip sounded slowed down	Used Set Rate in Audacity, which only changes playback speed.	Use Resample to change the sample rate.
Errors in AI-generated code	Wrong component name; suggested LittleFS did not run on Windows, and PlatformIO does not flash filesystem images on upload.	Verified against documentation instead of trusting the code as-is.

3. Technical Notes: ESP32-S3

Specs

Item	Details
CPU	2 main processor cores, plus a small ultra-low-power co-processor
Internal ROM (384 KB)	On-chip, not programmable
Internal SRAM (512 KB)	Volatile memory the CPU can access quickly
RTC Memory (16 KB)	Static RAM that persists through deep sleep
Flash (16 MB)	External memory on the board
PSRAM (8 MB)	External memory on the board

Boot Control

- Strapping pins: GPIO0, GPIO3, GPIO45, GPIO46.
- During chip reset, the hardware samples and stores the voltage level of the strapping pins. After reset is released, they work as normal-function pins.
- Boot mode is determined by GPIO0 and GPIO46 at reset. Modes: SPI Boot, Joint Download Boot, SPI Download Boot.
- Wi-Fi and Bluetooth LE only work when CPU_CLK uses PLL_CLK as its clock source.

Restricted Pins (do not use)

Pins	Reason
GPIO26–32	Connected to SPI flash / PSRAM
GPIO33–37	Used by Octal PSRAM extension

GPIO Matrix

- A crossbar switch between the chip's internal peripherals (I2S, SPI, UART, etc.) and the physical GPIO pins.
- Almost any usable GPIO can be assigned to almost any peripheral signal, so I2S pins (BCLK, LRC/WS, DIN) are chosen in software rather than hardwired.

4. Technical Notes: Audio (MAX98357A & I2S)

MAX98357A

Takes I2S digital audio input, decodes it to analog, and amplifies it directly into a speaker.

Pin	Function
LRC	Left/Right Clock: indicates whether current data is for the left or right channel
BCLK	Bit clock for reading data on the data pin
DIN	Data In: carries both left and right channel data
GAIN	Voltage sets the gain. Connect to GND, leave floating, or connect to Vin (through resistors) to select gain; see datasheet table
SD	Voltage selects mode: shutdown, left, right, or (left + right)/2

I2S and PCM

- I2S is a digital audio protocol that carries PCM sample data as signed two's complement values.
- PCM samples the analog signal's amplitude at uniform intervals and quantizes each sample to the nearest digital step.
- Sample magnitude determines volume. 16-bit range: –32,768 to +32,767.
- Positive values push the speaker cone out; negative values pull it in. Rapid alternation vibrates the cone and produces sound.
- PCM values carry no frequency information on their own; frequency comes from how quickly the code feeds values to the I2S hardware.

5. Audio Clip Playback over Wi-Fi

Preparing Clips (Audacity)

- Record the clip, then export as signed 16-bit PCM WAV.
- Change sample rate with **Resample**, not Set Rate (Set Rate only slows the audio down).
- Mix stereo down to mono.

Playback Setup

- Set the ESP32-S3 to SoftAP mode so it creates its own Wi-Fi network.
- Run a small web server: tapping a button sends a request that plays the matching clip.
- Connect a phone to the board's Wi-Fi and open the board's IP in a browser.

6. Circuit Fundamentals

- Picture all VCC lines as one conductive plate: they are all connected, so if one point is at 5 V, every point is at 5 V. There is no “input” or “output” end.
- Picture all GND lines as a second plate that is always at 0 V.

7. Lessons Learned

- For driver and Windows issues: search the problem, understand it, then reason through the fix rather than blindly following AI instructions.
- Use Claude as a helper, but verify its output; it produced incorrect component names and unworkable suggestions.
- Check physical connections (wire gauge, solder joints) early when hardware is silent.

8. Next Steps

- Understand the playback and web server code to avoid depending on Claude.
- Build the device frame.
- Mount the device on the dog.